VIGNESH AJITH NAIR

Data Engineer | Machine Learning
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SUMMARY

M.S. Computer Science candidate (GPA 3.78, graduating Dec 2026) who turns messy, unstructured data into clean, validated, analysis-ready datasets and ships them into real products. Built a field-image pipeline and classifier now deployed in a community reporting platform, a multi-stage document-to-structured-data pipeline, and a RAG evaluation harness. Google Cloud and AWS certified, published at Springer ICCIDA, with enterprise Agile experience at SAP Labs.

EDUCATION

Georgia State University Atlanta, GA | Expected Dec 2026
M.S. Computer Science | GPA 3.78/4.0
Certifications: Google Associate Cloud Engineer (2024) • AWS Certified Cloud Practitioner (2025)

TECHNICAL SKILLS

Data Engineering: Python (pandas, NumPy, PySpark), SQL, Apache Spark, Databricks, Apache Airflow, ETL/ELT pipelines, data modeling, data quality & validation
Databases & Cloud: PostgreSQL, MySQL, SQLite, MongoDB | AWS (EC2, S3), Google Cloud Platform, Docker, Linux, Git
Machine Learning & AI: PyTorch, TensorFlow, scikit-learn, XGBoost, Hugging Face, embeddings, RAG, LLM evaluation
Analytics & Backend: Tableau, Power BI, Jupyter | FastAPI, REST APIs, Spring Boot, React | Java, JavaScript, C++

EXPERIENCE

Graduate Research Assistant — *Georgia State University* Atlanta, GA | Jan 2025 – Present
- **Shipped to production:** integrated a sanitation-hazard image classifier into the MaisLugar community reporting platform, replacing manual review of resident-submitted photos.
- Co-built the end-to-end data pipeline (collection, labeling, cleaning, preprocessing) with research partners across 4 neighborhoods in Salvador, Brazil, turning inconsistent field photos into a single model-ready dataset.
- Trained a binary classifier (exposed plumbing vs. open sewage) to 76% test accuracy from only 104 labeled images, holding consistent across all 4 neighborhoods despite varied angle, distance, and lighting; extended to a 5-class model (65% on 167 images).

Software Intern — *SAP Labs* Bengaluru, India | Aug 2024 – Dec 2024
- Delivered enhancements to an enterprise SAP application across 10+ Agile sprints, tracking features and production defects in Jira and validating every fix before release.
- Diagnosed production defects alongside senior engineers by tracing behavior through unfamiliar code, application logs, and documentation to identify root causes.

Research Assistant — *Ammachi Labs* Kerala, India | Sep 2023 – Aug 2024
- Built reproducible Python workflows for data preparation, experimentation, and model evaluation on computer vision and ML projects, with documented datasets and design decisions so teammates could rerun and extend experiments.

PROJECTS & PUBLICATION

ConflictGuard AI — M.S. Capstone | *Python, FastAPI, RAG, Agentic AI* Aug 2026 – Present
- Architecting a multi-stage RAG pipeline (retrieval → claim extraction → conflict detection → source evaluation → verification) that flags contradictory evidence across sources.
- Designing an evaluation harness that scores accuracy, faithfulness, citation correctness, unsupported-claim rate, and latency to compare pipeline configurations objectively.

Semantic Classification & Clustering of Print Media | *Python, Embeddings, React* Jan 2026 – Apr 2026
- Built a multi-stage document pipeline (ingestion, classification, layout-aware extraction, embeddings, clustering) converting unstructured print media into structured records, with a React 2D/3D explorer for analysis.

Food Runner — Full-Stack Delivery Platform | *SQL, JavaScript* Aug 2025 – Dec 2025
- Designed the relational SQL data model for a multi-role app (customers, restaurants, drivers) covering role-based access, ordering, driver assignment, and order-lifecycle tracking.

Micro Facial Expression Recognition | *Python, XGBoost, GA-PSO* Published, Springer ICCIDA 2024
- Engineered features from the CASME II dataset and paired XGBoost with hybrid GA-PSO feature selection to handle subtle motion, noise, and class imbalance.